Prasant Koirala

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Education

The University of Southern Mississippi — Hattiesburg, MS
Bachelor of Science in Computer Science, GPA: 3.956 — *Expected, May 2028*
Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Machine Learning

Experience

Production Engineering Fellow — June 2026
Meta — *Remote USA*
- Selected from a competitive applicant pool to be one of 100 Production Engineering Fellows hosted by MLH in collaboration with Meta.

AI/ML Research Assistant — Sept. 2025 – Feb. 2026
The University of Southern Mississippi — *Hattiesburg, MS*
- Implemented a ModernBERT-based co-attention architecture for social engineering detection, modeling subject–body interactions and achieving 100% test accuracy on 158 email pairs.
- Built dataset construction and preprocessing workflows from a preprocessed corpus derived from 500,000+ corporate emails, generating balanced sender–receiver pairs across 4 attacker identities.

Web/App Development Intern — Jun. 2025 – Aug. 2025
Optimal Answers, LLC — *Gulfport, MS*
- Built and shipped backend features for internal and customer-facing web applications using C#, SQL, and REST APIs, supporting day-to-day workflows for business and operations teams.
- Refactored performance-critical backend logic and optimized database queries, reducing UI latency and cutting employee task time by ∼30% across core application flows.
- Designed and deployed a staging and pre-release validation environment, enabling safe testing and reducing production incidents by ∼40%.

Projects

Lavender – Open Source EDR | ***Go, Rust, Distributed Systems*** — GitHub
- Designed and built a fully functional open-source EDR system built for easy deployment across large distributed environments.
- Built an eBPF kernel module in Rust for system-level event capture and developed supporting infra services in Go.
- Implemented a Canonical Event Stream over NATS as a single source of truth, ensuring consistent data flow across detection workers, the dashboard, and the database.

MangOS: x86 Operating System | ***C, x86, Systems Programming*** — GitHub
- Implemented a custom x86 Operating System bootstrapped via the Limine bootloader, transitioning from bootloader state into protected kernel execution with a team of 6 developers.
- Built a kernel-level framebuffer terminal with direct video memory access, supporting text rendering, scrolling, and cursor control without libc.
- Designed low-level memory and I/O abstractions to safely interface with hardware from kernel space.

Hirely - AI Agent For Hiring | ***IBM WatsonX Hackathon*** — GitHub
- Built a multi-AI agent system using IBM watsonx Orchestrate that ingests GitLab and Jira contribution data, runs automated agentic workflows, and aggregates structured employee skill profiles into Cloudant via OpenAPI tools
- Implemented vectorization and semantic search using FAISS, enabling vector search that matches natural-language job descriptions to internal candidates based on real contribution history rather than resumes
- Integrated Dockerized services and OpenAPI-exposed tools to allow agents to perform candidate search and profile retrieval, supporting scalable internal hiring queries through embedding-based similarity ranking

Technical Skills

Languages: Go, C++, Python, Rust, TypeScript
Frameworks: FastAPI, Node.js (Express), React, TailwindCSS
Systems & Infra: Git, Docker, Google Cloud, AWS
Databases: PostgreSQL, MongoDB
AI/ML: Pytorch, Pandas, Numpy, TensorFlow, FAISS, LLM APIs (embeddings)